

*My interest in AI/ML basically started with one question I couldn't shake: how do you actually get a machine to learn something useful from data, not just spit out a number? I'm doing Mechanical Engineering at Jadavpur University, so none of this was part of my syllabus — I picked it up on the side, mostly by building stuff and figuring out why it broke.*

*First real project was a concrete compressive strength predictor, a Scikit-learn pipeline taking material composition and curing age and predicting strength at around 91% accuracy. Sounds basic, but that's the project where feature scaling and train-test splits stopped being textbook terms and became things I'd actually messed up and had to fix. After that came a solar collector efficiency predictor — Linear Regression, Random Forest, XGBoost, feature importance, and a Streamlit interface so I could actually poke at it instead of reading numbers off a screen.*

*Then I got into generative AI. I built PromptCraft, a prompt optimization tool on Llama 3.3 via the Groq API. This is where "the model works" and "the system works" stopped meaning the same thing — prompt engineering, backend wiring, weird inputs breaking things in ways I hadn't planned for.*

*Right now I'm part of a team building an AI assistant robot, and I'm handling the RAG pipeline — PDF ingestion, semantic chunking, sentence-transformer embeddings, ChromaDB, FastAPI, PostgreSQL. Out of everything, this has been the hardest to get right. Chunking took several tries — too small and the context disappears, too large and retrieval gets messy. I spent a frustrating few days figuring out why retrieval kept returning irrelevant chunks, and that one bug taught me more about how embeddings actually behave than anything I'd read.*

*What I don't have is the theory underneath all of this. I can make a pipeline work, but when it breaks in a weird way, I'm mostly guessing and adjusting rather than reasoning through it — optimization, probability, why some architectures generalize and others don't. I'm also working on a research project on thermoacoustic instability in long ducts, and I keep wondering if a data-driven angle could complement the analytical side of it, but I don't have the grounding to actually try that yet.*

*That's the gap I want MLSS to close. Mechanical engineering gave me comfort with physical systems, but honestly the bigger thing is that my curiosity never really stayed inside one department — every time I hit a wall, I just went and learned whatever was on the other side, whether that meant ML, backend systems, or vector databases. I'm not here to add a line to my resume. I've reached the point where going further means actually understanding why things work, not just getting them to.*